Arush Bodla

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Education

University of Maryland – Banneker/Key Scholar College Park, MD
B.S. in Computer Science (Cybersecurity) *Aug. 2026 – Present*

Academies of Loudoun Leesburg, VA
Academy of Engineering & Technology, Computer Science Pathway — GPA: 4.0/4.0 *Aug. 2022 – June 2026*

Experience

ManTech — DFEND Program Herndon, VA
Digital Forensics Intern *June 2026 – Aug. 2026*

- Reverse-engineered stripped, self-modifying Linux binaries with GDB's Python API and Unicorn CPU emulation to defeat XOR-encrypted anti-debugging checks and brute-force segmented keys
- Performed an Android app security assessment (APKTool, JADX, ADB, Frida) using SMALI-level code injection and runtime instrumentation to bypass client-side security controls
- Reverse-engineered a GPS-tracker's companion app, tracing device telemetry over RTSP to attribute the product to a shell-company supply-chain risk

ACL Competitive Programming Club Leesburg, VA
Co-Founder & Tournament Director *Sept. 2023 – June 2026*

- Architected and deployed a self-hosted DOMjudge platform on 3 AWS EC2 instances, isolating judgehosts in a VPC/NAT gateway and load-balancing to sustain 500+ submissions in a live 3-hour window
- Scaled EC2 capacity under live load, built custom AMIs for judgehost/web-server redundancy, and resolved a production leaderboard failure in real time on contest day
- Directed a 100+ competitor tournament on a $3,900 budget; migrated 13 problems from Polygon to DOMjudge via pol2dom, authoring custom checkers and test-data generators

John Champe CyberPatriot Aldie, VA
Linux Lead & Co-Treasurer *Aug. 2022 – June 2026*

- Built a 4,000-line, 22-module Bash hardening tool (SSH, PAM, firewall, kernel, Apache/MySQL/PHP), contributing to an All-Virginia 1st place finish and 13th/4,786 nationally
- Owned incident response on compromised Linux VMs as Linux Lead — forensic artifact removal, user auditing, and config hardening under time pressure
- Managed a $3,000+ budget as Co-Treasurer and led weekly training on system hardening, network defense, and CTF strategy, growing participation from 15 to 75 students

Projects

ACL-OJ | *Next.js, TypeScript, Supabase, Docker, Judge0* Aug. 2025 – Present

- Architected a full-stack judge (Next.js/TypeScript, Supabase/PostgreSQL, Dockerized Judge0) returning real-time verdicts for Python, C++, and Java submissions
- Designed a LAN-deployable, fully offline architecture with Docker Compose, eliminating reliance on AI-detection tooling
- Built a Codeforces-style rating engine, live leaderboards, and an access-controlled admin CRUD panel on a schema designed from scratch

RegiMon | *Rust, React, Tauri, Win32 API, GitHub Actions* Aug. 2025 – Present

- Built a real-time Windows registry monitor in Rust using the Win32 `RegNotifyChangeKeyValue` API to detect malware persistence with zero polling overhead
- Implemented recursive change tracking, one-click rollback, and privilege-aware admin elevation; validated the tool by detecting and mitigating the LoveBug virus
- Curated a security-relevant default key set (autorun, browser-hijack, credential store) and automated build/release with GitHub Actions CI/CD

Awards & Honors

- **National Cyber League** — 4th Place Nationally, High School Division (2025 Spring & Fall)
- **IIOT Team USA** — 1 of 4 students selected nationally for the International Informatics Olympiad in Teams
- **Competitive Programming** — **USACO Gold**, 1st place Philadelphia Classic, WWPIT

Technical Skills

Languages: C++, Java, Python, C#, TypeScript, JavaScript, Bash, Rust
Backend & Infra: AWS (EC2, VPC), Docker, Supabase/PostgreSQL, DOMjudge, Judge0, GitHub Actions, Next.js
Security & Certs: Ghidra, GDB, Unicorn, Frida, APKTool, JADX, ADB, CTF/RE, System Hardening, Security+, AWS CCP, SC-900